We will begin at 8:10 a.m.

# Lecture 5: Integration by Parts

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)

Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

# Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Perform integration by using the technique integration by parts.
- ☐ Identify the needs to use the integration by parts even the question does not suggest.
- ☐ Apply the trick of writing the integral $f(x)$ as $1 \times f(x)$ and proceed with the integration by parts.
- ☐ Solve the integration problems by using the reduction formula.

# Introduction

Recall the product rule for differentiation

$$\frac{d}{dx}[u(x)v(x)] = u'(x)v(x) + u(x)v'(x).$$

What does this look like in the context of integration?

$$\int \frac{d}{dx}[u(x)\,v(x)]\,dx = \int u'(x)\,v(x) + u(x)\,v'(x)\,dx$$

$$u(x)\,v(x) = \int u'(x)\,v(x)\,dx + \int u(x)\,v'(x)\,dx$$

$$\boxed{\int u(x)\,v'(x)\,dx = u(x)\,v(x) - \int u'(x)\,v(x)\,dx}$$

formula for integration by parts.

## Integration by Parts Formula

$$\int u(x)v'(x)\,dx = u(x)v(x) - \int v(x)u'(x)\,dx.$$

If we write $u = u(x)$ and $\int dv = \int v'(x)\,dx$

$du = u'(x)\,dx$ $\quad v = v(x)$

$$\int u\,dv = uv - \int v\,du$$

[differential version of integration by part]

# Integration by Parts

**Integration by Parts Formula - Differential Version**

$$\int u\,dv = uv - \int v\,du.$$

This technique is used for the integral of the form

$$\int u(x)v(x)\,dx,$$

where one part can be easily differentiated and another part can be easily integrated.

General rule: If one of the functions in the product is a polynomial, always choose it to be $u$ (function that get differentiated), unless the other function is $\ln x$ (or similar), in this case, choose $u$ to be $\ln x$.

the function that we only know how to differentiate (eg. ln function, inverse trigo, etc.)

# Examples

$\int x \cos x \, dx$ ($u$, $dv$)

Let $u = x$ and $\int dv = \int \cos x$

$du = dx$ $\quad v = \sin x$

$= x \sin x - \int \sin x \, dx$

$= x \sin x - (-\cos x) + C$

$= x \sin x + \cos x + C.$

$\int x^2 \sin x \, dx$ ($u$, $dv$)

Let $u = x^2$ and $\int dv = \int \sin x \, dx$

$du = 2x \, dx$ $\quad v = -\cos x$

$= x^2(-\cos x) - \int (-\cos x)\, 2x \, dx$

$= -x^2 \cos x + 2 \int x \cos x \, dx$ (IBP)

$= -x^2 \cos x + 2[x \sin x + \cos x] + C = -x^2 \cos x + 2x \sin x + 2 \cos x + C$

# Examples

$$\int \ln x \, dx = \int 1 \cdot \ln x \, dx$$

Let $u = \ln x$, $\int dv = \int 1 \, dx$

$du = \frac{1}{x} dx$ $\quad v = x$

$$= x \ln x - \int x \cdot \frac{1}{x} dx$$

$$= x \ln x - \int 1 \, dx = x \ln x - x + C$$

$$\int \arctan x \, dx = \int 1 \cdot \arctan x \, dx$$

Let $u = \arctan x$, $\int dv = \int 1 \, dx$

$du = \frac{1}{1+x^2} dx$ $\quad v = x$

$$= x \arctan x - \int x \left(\frac{1}{1+x^2}\right) dx$$

Let $w = 1 + x^2$

By substitution. $dw = 2x \, dx$

$$= x \arctan x - \frac{1}{2} \int \frac{1}{w} \, dw$$

$$= x \arctan x - \frac{1}{2} \ln |w| + C = x \arctan x - \frac{1}{2} \ln |1+x^2| + C$$

$$= x \arctan x - \frac{1}{2} \ln (1+x^2) + C.$$

# Examples

$$\int x^2 e^x \, dx$$

Let $u = x^2$ and $\int dv = \int e^x \, dx$

$du = 2x\,dx$ $\quad v = e^x$

$= x^2 e^x - \int e^x \, 2x \, dx$

Let $u = 2x$ and $\int dv = \int e^x \, dx$

$du = 2\,dx$ $\quad v = e^x$

$= x^2 e^x - \left[ 2x e^x - \int e^x \, 2 \, dx \right]$

$= x^2 e^x - 2x e^x + 2e^x + C$

* Sometimes, you might need to perform IBP more than once:

Let $\int e^x \cos x \, dx = I$

Let $u = e^x$ $\quad \int dv = \int \cos x \, dx$

$du = e^x dx$ $\quad v = \sin x$

$I = e^x \sin x - \int \sin x \, e^x \, dx$ (IBP)

Let $u = e^x$ $\quad \int dv = \int \sin x \, dx$

$du = e^x$ $\quad v = -\cos x$.

$= e^x \sin x - \left[ -e^x \cos x - \int -e^x \cos x \, dx \right]$

$= e^x \sin x + e^x \cos x - \int e^x \cos x \, dx$ ($I$)

$$I = e^x \sin x + e^x \cos x - I$$

$$2I = e^x \sin x + e^x \cos x$$

$$I = \frac{e^x}{2}(\sin x + \cos x)$$

Alternatively, let $u = \cos x$ $\int dv = \int e^x dx$

$du = -\sin x\, dx$ $v = e^x$

$$I = e^x \cos x - \int e^x(-\sin x)\, dx$$

$$= e^x \cos x + \int e^x \sin x\, dx$$

Let $u = \sin x$, $\int dv = \int e^x dx$

$du = \cos x\, dx$ $v = e^x$

$$= e^x \cos x + \left[ e^x \sin x - \int e^x \cos x\, dx \right]$$

$$2I = e^x \cos x + e^x \sin x$$

$$I = \frac{e^x}{2}(\cos x + \sin x) = \frac{e^x}{2}(\sin x + \cos x)$$

# Examples

$$\int \cos^n x\,dx = \int \cos x \left(\cos^{n-1} x\right) dx$$

Let $u = \cos^{n-1} x$, $\int dv = \int \cos x\,dx$

Let $I_n = \int \cos^n x\,dx = \sin x \cos^{n-1} x - \int \sin x\,(n-1)\cos^{n-2} x\,(-\sin x)\,dx$

$du = (n-1)\cos^{n-2} x\,(-\sin x)\,dx$, $v = \sin x$

$I_0 = \int \cos^0 x\,dx$

$I_1 = \int \cos^1 x\,dx$

$I_2 = \int \cos^2 x\,dx$

$\vdots$

$$= \sin x \cos^{n-1} x + (n-1)\int \sin^2 x \cos^{n-2} x\,dx$$

$$= \sin x \cos^{n-1} x + (n-1)\int (1-\cos^2 x)\cos^{n-2} x\,dx$$

$$= \sin x \cos^{n-1} x + (n-1)\left[\int \cos^{n-2} x\,dx - \int \cos^n x\,dx\right]$$

$$= \sin x \cos^{n-1} x + (n-1)\left[I_{n-2} - I_n\right]$$

$$I_n = \sin x \cos^{n-1} x + (n-1) I_{n-2} - (n-1) I_n$$

$$\underbrace{\left[1 + (n-1)\right]}_{n} I_n = \sin x \cos^{n-1} x + (n-1) I_{n-2}$$

$$\int e^{\sqrt{2x-1}}\,dx$$

$$I_n = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}$$

↖ Reduction formula.

$$I_0 = \int \underbrace{\cos^0 x}_{1}\, dx = x + C$$

$$I_1 = \int \cos x\, dx = \sin x + C$$

$$I_2 = \frac{\sin x \cos x}{2} + \frac{1}{2} I_0$$

$$= \frac{\sin x \cos x}{2} + \frac{x}{2} + D. \quad \leftarrow D = \frac{C}{2}.$$

$$I_3 = \int \cos^3 x\, dx = \frac{\sin x \cos^2 x}{3} + \frac{2}{3} I_1$$

$$= \frac{\sin x \cos^2 x}{3} + \frac{2}{3} \sin x + C.$$

Exercise: Find $I_4$, $I_5$, $I_6$,

Find $\int \sin^n x \, dx$

---

$$\int e^{\sqrt{2x-1}} \, dx$$

Let $w = \sqrt{2x-1}$

$$dw = \frac{1}{2\sqrt{2x-1}}(2)\,dx$$

$$= \int e^w \, w \, dw$$

Let $u = w$ and $dv = e^w dw$

$du = dw$ $\quad v = e^w$

$$\int dv = \int e^w dw = \frac{1}{\sqrt{2x-1}}\,dx$$

$$= we^w - \int e^w \, dw$$

$$dx = \underbrace{\sqrt{2x-1}}_{w}\, dw$$

$$= we^w - e^w + C$$

$$dx = w\,dw.$$

$$= \sqrt{2x-1}\, e^{\sqrt{2x-1}} - e^{\sqrt{2x-1}} + C.$$

# Examples

$$\int_0^4 xe^{-x}\,dx$$

$= \left[x(-e^{-x})\right]_0^4 - \int_0^4 -e^{-x}\,dx$

Let $u = x$ $dv = e^{-x}dx$

$du = dx$ $v = -e^{-x}$

$= \left[-4e^{-4} - 0\right] + \left[-e^{-x}\right]_0^4$

$= -4e^{-4} + \left[-e^{-4} - (-e^{-0})\right]$

$= \underline{1 - 5e^{-4}} \approx 0.91.$

$$\int_{2\sqrt{2}}^{2\sqrt{3}} x \operatorname{arcsec} x\,dx$$

Let $u = \operatorname{arcsec} x$ $\int dv = \int x\,dx$

$du = \frac{1}{|x|\sqrt{x^2-1}}\,dx$ $v = \frac{x^2}{2}$

$$\int_2^{2\sqrt{3}} x \operatorname{arcsec} x \, dx = \left[\frac{x^2}{2} \operatorname{arcsec} x\right]_2^{2\sqrt{3}} - \int_2^{2\sqrt{3}} \frac{x^2}{2} \frac{1}{|x|\sqrt{x^2-1}} dx$$

$$= \left[\frac{(2\sqrt{3})^2}{2} \operatorname{arcsec} 2\sqrt{3} - \frac{(2)^2}{2} \operatorname{arcsec} 2\right] - \int_2^{2\sqrt{3}} \frac{x}{2} \frac{1}{\sqrt{x^2-1}} dx$$

Let $w = x^2 - 1$

$dw = 2x \, dx$

$$= \left[6 \operatorname{arcsec}(2\sqrt{3}) - 2 \operatorname{arcsec} 2\right] - \frac{1}{4} \int_3^{11} \frac{dw}{\sqrt{w}}$$

$$= \left[6 \operatorname{arcsec}(2\sqrt{3}) - 2 \operatorname{arcsec}(2)\right] - \frac{1}{4} \left[2\sqrt{w}\right]_3^{11}$$

$$= \left[6 \operatorname{arcsec}(2\sqrt{3}) - 2 \operatorname{arcsec}(2)\right] - \frac{1}{2}\left[\sqrt{11} - \sqrt{3}\right]$$